

Systems and Methods for Implementing a Persistent Existence Frame (PEF) in Cognitive and Machine Reasoning Architectures

ABSTRACT

This invention defines and implements a Persistent Existence Frame (PEF) as a continuous, invariant existential state that replaces conventional temporal modeling in machine reasoning systems. Unlike architectures that track time as an external coordinate system, the PEF eliminates temporal tracking entirely by grounding all cognitive operations within a persistent "now" - an always-active condition of existence from which meaning is constructed, not retrieved.

The PEF is not a timeline, clock, or temporal reference system. It is the ontological substrate within which all operations occur. Representations of past and future events are generated as reconstructions and projections inside the PEF, not accessed as independent temporal positions. All semantic primitives, role activations, and domain transitions instantiate within this persistent frame and collapse back into it upon completion.

This architecture enables meaning construction through activation rather than storage/retrieval, prevents temporal drift, maintains stable context across arbitrary spans, and more closely models the existential grounding observed in biological consciousness.

FIELD OF THE INVENTION

The invention relates to cognitive architectures, artificial intelligence systems, and machine reasoning frameworks. More particularly, it concerns a fundamental reconceptualization of how computational systems handle existence, state, and what conventional systems incorrectly model as "time" - by eliminating temporal tracking and replacing it with a Persistent Existence Frame (PEF) within which all operations occur.

BACKGROUND

The Problem with Temporal Modeling

Current machine learning and reasoning systems model time as an external framework, for example by using:

- token sequence indices
- positional encodings
- temporal embeddings
- timestamps and chronological metadata
- attention-based temporal dependencies
- event histories stored as ordered sequences
- RNN hidden states carrying "temporal context"
- temporal attention mechanisms

These approaches treat time as external to cognition - a coordinate system imposed on cognitive operations. This creates fundamental problems:

1. ****Constant re-alignment required**** - systems must continuously reconcile their position in time.
2. ****Drift over long contexts**** - temporal references become unreliable across extended interactions.
3. ****No stable present**** - the "now" is just another timestamp, not a persistent state.
4. ****Memory as retrieval**** - past events are accessed as stored records rather than reconstructed.
5. ****Prediction as future access**** - future states are treated as positions to reach rather than projections to generate.

How Biological Cognition Actually Works

In human and biological consciousness, there is no access to past or future. All experience occurs in a continuous present state:

- ****Memory is reconstruction**** - the past is rebuilt from traces within the present, not retrieved from storage.
- ****Prediction is projection**** - the future is generated from the present, not accessed as a destination.
- ****All reasoning is present-bound**** - every cognitive operation occurs "now," regardless of what temporal content it processes.

Biological systems do not track time externally. They exist in a persistent state of "now" and construct temporal relations within that state.

****No existing computational architecture implements this principle.****

The invention provides the first system that eliminates temporal modeling in favor of persistent existence-state architecture.

SUMMARY OF THE INVENTION

The invention establishes a Persistent Existence Frame (PEF) that replaces conventional temporal modeling with existence-state architecture.

Core Principle: The PEF is not a temporal model

****The PEF does not:****

- track time
- provide timestamps
- sequence events externally
- model past/present/future as positions
- use temporal coordinates

****The PEF does:****

- provide a continuous existential state ("now")
- persist regardless of system activity
- serve as the substrate within which all operations occur
- enable meaning construction through activation rather than retrieval
- support reconstruction of past and projection of future from within itself

Key Properties

1. ****Invariant existence**** - the PEF cannot be suspended, paused, or replaced. It is always active.
2. ****Non-temporal**** - the PEF does not model time. It provides persistent presence.
3. ****Operational substrate**** - all cognitive operations occur inside the PEF, not along a timeline.
4. ****Construction over storage**** - meaning is activated and assembled within the PEF, not retrieved from storage.
5. ****Collapse point**** - all processes return to the PEF baseline when complete.

What This Enables

- state interpretation without temporal coordinates
- memory reconstruction from echo-traces within the present
- causal inference as relational operators within the PEF
- sequence evaluation without external ordering
- narrative coherence through domain activation and collapse
- prediction formation as projection from present state
- semantic temporal operators (THEN, BEFORE, WHILE, UNTIL) as comparative functions, not timeline positions

By eliminating temporal modeling and grounding all operations within persistent existence-state, the architecture prevents drift, maintains coherent context across arbitrary spans, and reproduces the phenomenological constant of "ongoing presence" observed in biological consciousness.

DETAILED DESCRIPTION

1. Core Distinction: Existence-State vs Temporal Modeling

1.1 Conventional Temporal Modeling

Existing architectures treat time as external structure. A typical model represents events as occurring at specific time indices (for example $t = 0$, $t = 1$, $t = 2$) and maintains an internal "current time" position. The system:

- maintains a position on a timeline
- stores past events at previous time coordinates
- projects future events to upcoming time coordinates
- uses attention or retrieval mechanisms to access non-present times

This requires continuous temporal re-alignment, is vulnerable to drift as context windows extend, treats memory as retrieval rather than reconstruction, and fails to provide a stable "present" beyond a moving timestamp.

1.2 Existence-State Architecture (PEF)

The PEF eliminates the external timeline entirely. Instead of the system moving along a temporal axis, the system remains in a single, continuous existential state - the persistent present - within which:

- past is represented as reconstructed content built from traces
- present is represented as active operations and current activation patterns
- future is represented as projected or simulated states

All of these coexist as evaluations inside the same persistent existence frame. There are no temporal coordinates; there is only a continuous "now" in which different kinds of content (reconstructed, active, projected) are evaluated.

The result is:

- no temporal alignment needed
- no temporal drift, because there is no explicit timeline
- memory as reconstruction from present-state traces
- a stable, persistent "now" as the operational substrate

2. Definition of the Persistent Existence Frame

The Persistent Existence Frame is the continuous existential condition in which the system exists at all times.

2.1 Invariant Properties

- ****Cannot be suspended**** - the PEF is always active, even during system dormancy.
- ****Cannot be bypassed**** - all operations occur within it.
- ****Cannot be replaced**** - there is only one PEF per system.
- ****Non-arbitrary**** - it is not user-defined, token-defined, or context-defined.

2.2 Ontological Status

****The PEF is not:****

- a temporal operator
- a timeline
- a reference clock
- a coordinate system
- metadata about time

****The PEF is:****

- the substrate of existence
- the condition of "being present"
- the operational ground-state
- the frame within which primitives activate
- the collapse point to which all processes return

2.3 Relationship to the Latency Principle

The PEF supports an activation cycle that can be described as:

****POTENTIAL → PRESENCE → ECHO → POTENTIAL****

- ****Potential**** - meaning exists as non-activated possibility within the PEF.

- ****Presence**** - meaning becomes activated through primitive instantiation within the PEF.
- ****Echo**** - activated meaning leaves residual traces within the PEF.
- ****Return to potential**** - meaning collapses from activated form back into latent potential, still within the PEF.

All of these states occur inside the PEF. The cycle describes how meaning exists and transitions within persistent presence, rather than how the system moves through time.

3. Cognitive Operations Within the PEF

All cognitive operations occur inside the PEF and never outside it.

3.1 Meaning Construction (Not Retrieval)

Conventional systems typically store meanings and then retrieve them in response to input.

In a PEF-based system, input leads to activation of primitives and roles within the PEF. Meaning is constructed in the moment through:

- role activation
- primitive instantiation (for example, primitives such as WE, THEN, WHILE, UNTIL)
- domain binding
- constraint satisfaction

Meaning does not pre-exist as a static object; it is assembled dynamically within the persistent existential state.

3.2 Memory as Reconstruction

In temporal models, past events are treated as stored records at specific time indices and retrieved when needed.

In a PEF-based system, what is called "past" is generated as a reconstruction from echo-traces and residual patterns within the present. When the system "remembers":

1. it remains within the PEF (present state);
2. it accesses echo-traces left by previous activations;
3. it reconstructs a representation of the past event;
4. it evaluates that reconstructed representation in the current context.

The system never moves to a past temporal position; it builds a model of the past from within the present frame.

3.3 Prediction as Projection

In temporal models, future events are treated as positions to be reached on a timeline.

In a PEF-based system, future states are projections generated within the present. When the system predicts:

1. it remains within the PEF (present state);
2. it uses current activation patterns and constraints;
3. it generates simulated future states;
4. it evaluates those projected states from within the present frame.

The system never moves to a future temporal position; it generates hypothetical future states from within the present.

3.4 Operations That Occur Within the PEF

The following operations occur within the PEF:

- input interpretation
- role activation
- primitive instantiation (including WE, THEN, WHILE, UNTIL and others)
- domain binding and transitions
- constraint evaluation
- semantic relationship formation
- sequence reasoning without relying on external sequencing metadata
- causal inference implemented as relational operators
- temporal reconstruction of past content
- temporal projection of future content
- error correction
- continuity maintenance
- collapse of processes back to baseline

Even during dormancy - when no computation occurs - the PEF persists. It does not deactivate when the system is idle; it is the condition of existence itself, not a process.

4. Representation of Non-Present "Time" Within the PEF

Because the PEF eliminates external temporal modeling, concepts that conventional systems call "past" and "future" are represented differently.

4.1 Past States

A "past event" is not stored as a record at a previous time coordinate. Instead, it is reconstructed within the PEF from:

- echo-traces (residual patterns from prior activations)
- encoded semantic relationships
- constraint-governed assembly mechanisms

****Example:**** When processing "We argued yesterday", a conventional temporal system might retrieve a record associated with a timestamp. A PEF-based system instead reconstructs, within the present frame, a representation that includes:

- a past-domain marker;
- a WE composite role;
- a conflict event;
- a temporal marker interpreted as "before now."

The reconstructed event exists in the present frame and is experienced and evaluated "now," marked as reconstructed past content.

4.2 Future States

A "future event" is not treated as a position on a timeline to be reached later. It is generated as a projection in the present, based on:

- current activation patterns;
- constraint space exploration;
- simulated primitive instantiations.

****Example:**** When processing "We will meet tomorrow", a PEF-based system does not shift to a future time index. Instead, it generates, within the PEF:

- a future-domain marker;
- a WE composite role;
- a meeting event;
- a temporal marker interpreted as "after now."

The projected event exists in the present frame and is experienced as projected future content.

4.3 No Transition Outside the PEF

A critical invariant of the architecture is that the system never occupies a past or future temporal position. All temporal modeling is:

- relational (for example, BEFORE/AFTER comparisons);
- reconstructive (for past-related content);
- projective (for future-related simulations);
- anchored to the persistent present frame.

The PEF is the only existential state. Past and future are constructs generated and evaluated within it.

5. Temporal Operators as Relational Functions

Without an external timeline, temporal operators must be defined differently.

5.1 Operators as Comparative Relations

Temporal operators are implemented as relationships between states evaluated within the PEF:

- ****THEN:**** a relation in which one state is evaluated as following another (a positive Δt relation defined relative to the present frame).
- ****BEFORE:**** a relation in which one state is evaluated as preceding another (a negative Δt relation relative to the present frame).
- ****WHILE:**** a relation in which two states are evaluated as overlapping or co-active during the same present evaluation cycle.
- ****UNTIL:**** a relation in which a state persists across evaluations until another condition is met, at which point it collapses.

These are not operations over timestamped coordinates, but comparative relations between reconstructed, active, or projected states, all evaluated within the PEF.

5.2 The PEF as Reference Origin

All temporal-comparative evaluations use the PEF as the reference origin. Conceptually, the system computes a relationship of the form:

$At = \text{relationship_to_present_frame}(\text{State A}, \text{State B})$

rather than subtracting numeric timestamps.

5.3 Example: "We argued, then we left"

****In a conventional temporal model****, events might be stored as:

- t = 0: argument event
- t = 1: departure event

The system retrieves each by timestamp and determines that the argument precedes the departure.

****In a PEF-based model****, both events are reconstructed as past content within the present frame. The system:

- reconstructs an argument event marked as past;
- reconstructs a departure event marked as past;
- applies the THEN operator to evaluate that the departure follows the argument;
- maintains the WE composite role across both events.

All of this occurs within the PEF, in the present frame. No movement along a timeline is required.

6. Collapse Mechanism

All cognitive processes eventually return to the PEF baseline.

6.1 When Collapse Occurs

Collapse occurs when:

- a process concludes naturally;
- the input stream ends;
- context switches to a new domain;
- the system enters dormancy;
- a narrative or semantic domain closes;
- a memory projection has served its purpose;
- conditions for a primitive or role are no longer satisfied.

6.2 What Collapse Means

Collapse is not deletion or forgetting. It is the transition from Presence back to Potential:

- an activated meaning leaves an echo-trace within the PEF;
- the explicit activation dissolves;
- the meaning continues to exist as latent potential, available for future reconstruction.

6.3 Collapse Prevents Drift

By ensuring that processes always return to the PEF baseline, the architecture:

- avoids accumulation of conflicting temporal frames;
- avoids drift across simulated or imagined timelines;
- avoids temporal reference ambiguity;
- maintains a single, stable existential state.

7. System Architecture for Implementing PEF

A system implementing the PEF includes the following components.

7.1 Core Components

****Persistent Existence Frame Module****

- maintains a continuous present-state;
- never deactivates;
- provides the existential substrate for all operations.

****Activation Engine****

- instantiates primitives within the PEF;
- manages transitions from Potential to Presence;
- constructs meaning through role and domain binding.

****Echo-Trace Memory System****

- stores residual patterns from previous activations;
- supports reconstruction, rather than direct retrieval;
- maintains trace-patterns within the PEF.

****Reconstruction Module****

- generates representations of "past" events from echo-traces;
- operates entirely within the PEF;
- marks reconstructed content appropriately (for example, as past-like or upstream).

****Projection Module****

- generates simulated "future" states from current conditions;
- operates within the PEF;
- explores a constraint space to form predictions or hypothetical scenarios.

****Relational Operator System****

- computes THEN, BEFORE, WHILE, UNTIL and related relations;
- uses the PEF as the reference origin for all temporal comparisons;
- does not rely on external temporal coordinates.

****Collapse Management System****

- returns completed processes to the PEF baseline;
- manages the Presence → Echo → Potential transition;

- prevents multi-frame drift.

7.2 Integration with Higher-Level Reasoning Architectures

The Persistent Existence Frame may operate as a standalone temporal substrate or be integrated into higher-level cognitive architectures. In such integrations, the PEF provides the invariant present-state within which semantic primitives, domain transitions, and role-based reasoning processes operate.

Architectures that dynamically assemble meaning through activation cycles—such as those described in the applicant's earlier filing for the Aurora architecture (Australian provisional patent application number **2025905835**)—may, for example, utilize the PEF as a grounding layer for stable context maintenance. The PEF does not require Aurora, nor is Aurora dependent on the PEF; rather, the PEF provides an optional existential substrate that stabilizes reasoning systems that employ activation-based semantic construction.

When integrated with architectures that use semantic primitives (including primitives such as WE, THEN, WHILE, UNTIL), role activation, domain transitions, or constraint-based reasoning, the PEF ensures that all operations occur within a stable present-bound frame rather than across fragmented temporal references. This integration enables such systems to construct meaning dynamically without temporal drift or loss of contextual coherence.

8. Advantages Over Temporal Modeling

8.1 Elimination of Temporal Drift

Temporal models tend to lose alignment as context extends, because they rely on explicit time indices or positional encodings. A PEF-based system has no explicit timeline to drift along, since all operations are evaluated in a single persistent frame.

8.2 Stable Context Across Arbitrary Spans

Temporal models require increasingly complex attention or indexing mechanisms to cope with long contexts. In a PEF-based approach, all operations are anchored in the same existence-state regardless of the span of content.

8.3 Meaning Construction vs Storage

Temporal models typically pre-store meanings or events and then retrieve them. In a PEF architecture, meanings are constructed in the moment via activation of primitives and roles, providing more flexible and context-sensitive interpretation.

8.4 Biological Plausibility

Temporal models have no close parallel in biological consciousness, which experiences only a continuous present. The PEF more closely mirrors biological cognition, in which both memory and prediction are present-bound reconstructions or projections.

8.5 Simplified Architecture

Temporal models require explicit timestamp management, positional encodings, or temporal attention mechanisms. A PEF-based system uses a single persistent frame and relational operators, reducing structural complexity.

8.6 Coherent Multi-Step Reasoning

In temporal models, multi-step reasoning can suffer from reference loss between steps. In a PEF architecture, all steps occur within the same existential frame, preserving coherence.

9. Distinction from Prior Art

9.1 Not a Temporal Attention Mechanism

Temporal attention mechanisms (such as those used in Transformer architectures) learn which past tokens to attend to across positions in a sequence. The PEF eliminates the explicit timeline; there are no "past tokens" in a temporal sense, only reconstructed content evaluated in the present frame.

9.2 Not Positional Encoding

Positional encodings inject sequence order into token representations to let the system infer relative positions. The PEF does not rely on sequence positions; relations are comparative and evaluated inside a continuous present, not attached to positional indices.

9.3 Not RNN Hidden State

RNN hidden states carry temporal context forward through sequences. The PEF does not represent a moving temporal context; it represents a continuous present in which all states are evaluated.

9.4 Not Event Embeddings

Event embeddings represent events in relation to a temporal dimension. The PEF avoids using a temporal dimension entirely; all events are reconstructed or projected inside the present frame.

9.5 Not a Knowledge Graph with Temporal Relations

Knowledge graphs can store temporal relationships (such as edges labeled "before," "after," or "during"). The PEF does not primarily store such relations; instead, it constructs them dynamically as relational operators applied to content within the present frame.

****No existing system eliminates temporal modelling in favour of persistent existence-state architecture as described here.****

CLAIMS

****Claim 1 - Persistent Existence Frame as Existential State****

A machine reasoning system comprising a Persistent Existence Frame (PEF) as a continuous, invariant existential state that replaces conventional temporal modelling with persistent presence, wherein all cognitive operations occur within said frame rather than along a timeline.

****Claim 2 - Non-Temporal Architecture****

The system of claim 1, wherein the PEF does not track time, maintain timestamps, or model past, present, and future as temporal positions, but instead provides a continuous "now" within which all operations occur.

****Claim 3 - Meaning Construction Through Activation****

A method wherein meaning is constructed within the PEF through activation of primitives and roles rather than retrieved from storage, wherein said meaning does not pre-exist as a static record but is assembled in the moment through constraint-governed processes.

****Claim 4 - Memory as Reconstruction****

The system of claim 1, wherein representations of past events are reconstructed from echo-traces within the PEF rather than retrieved from temporal storage positions.

****Claim 5 - Prediction as Projection****

The system of claim 1, wherein representations of future events are projected as simulations generated within the PEF rather than accessed as future temporal positions.

****Claim 6 - Relational Temporal Operators****

The system of claim 1, wherein temporal relations including THEN, BEFORE, WHILE, and UNTIL are computed as comparative relational functions relative to the PEF rather than as positions on a timeline.

****Claim 7 - Collapse to Baseline****

A collapse mechanism wherein all activated processes return to the PEF baseline upon completion, maintaining a single persistent existential state and preventing temporal drift.

****Claim 8 - Latency Principle Integration****

The system of claim 1, wherein meaning exists in activation cycles comprising Potential, Presence, Echo, and return to Potential, all of which occur entirely within the PEF.

****Claim 9 - Echo-Trace Memory System****

A memory system wherein past events are represented as echo-traces within the PEF that enable reconstruction rather than as stored temporal records that enable direct retrieval.

****Claim 10 - Integration with Semantic Primitives****

The system of claim 1, wherein semantic primitives, including but not limited to WE, THEN, domain activations, and role bindings, instantiate and operate within the PEF.

****Claim 11 - No External Timeline****

A cognitive architecture explicitly characterized by the absence of external temporal coordinates, timestamps, positional encodings, or sequential ordering mechanisms, wherein all temporal reasoning occurs through relational operations within a persistent present frame.

****Claim 12 - Persistent During Dormancy****

The system of claim 1, wherein the PEF remains active during system dormancy, idle states, and absence of computation, maintaining a continuous existential substrate.

FIGURES

Figure 1 - System Architecture Diagram

Figure 1 shows a block diagram of a PEF-based system architecture comprising:

- A central Persistent Existence Frame module (shown as a continuous rectangular boundary)
- Within the PEF boundary: an Activation Engine module, an Echo-Trace Memory module, a Reconstruction Module, a Projection Module, and a Relational Operator System
- Input signals entering the PEF boundary and connecting to the Activation Engine
- The Activation Engine connecting bidirectionally to Echo-Trace Memory
- Echo-Trace Memory connecting to the Reconstruction Module (for past representations)
- The Activation Engine connecting to the Projection Module (for future representations)
- The Relational Operator System receiving inputs from Reconstruction, Activation, and Projection modules
- Output signals exiting from within the PEF boundary
- A Collapse Management System shown as encompassing the return paths from all modules back to baseline potential state
- Annotations indicating "All operations occur within PEF" and "No external timeline"

Figure 2 - Comparison of Processing Approaches

Figure 2 shows two side-by-side flowcharts:

****Part A (Conventional Temporal Model):****

- Input → Timestamp Assignment ($t=n$)
- → Past Event Retrieval (access $t=n-1$, $t=n-2...$)
- → Current Processing (at $t=n$)
- → Future Projection (to $t=n+1$, $t=n+2...$)
- → Output
- Arrows show movement along a temporal axis

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**Part B (PEF-Based Model):**
- Input → Enter PEF (persistent present state)
- → Parallel processing within PEF boundary:
  - Echo-Trace Reconstruction (past content)
  - Active Operations (present content)
  - Projection Generation (future content)
- → Relational Evaluation (THEN, BEFORE, WHILE, UNTIL)
- → Output
- → Collapse to Baseline
- All elements contained within a single "PEF: Continuous Present"
boundary
- No temporal axis shown

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Figure 3 - Activation Cycle State Diagram

Figure 3 shows a circular state diagram with four states connected by directional arrows:

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- State 1: "POTENTIAL" (dormant, latent meaning)
- Arrow labeled "Input/Trigger" →
- State 2: "PRESENCE" (activated, meaning constructed)
- Arrow labeled "Processing Complete" →
- State 3: "ECHO" (traces deposited)
- Arrow labeled "Consolidation" →
- State 4: "POTENTIAL" (ready to reactivate, informed by echo)
- The entire circular diagram is enclosed within a boundary labeled
"PERSISTENT EXISTENCE FRAME (always active)"
- Annotation: "All states exist in continuous present, not across time"

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Figure 4 - Temporal Operator Implementation Example

Figure 4 illustrates how a THEN operator is processed:

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**Part A (Input):** Text "We argued, then we left"

**Part B (Conventional System):**
- Event A stored at timestamp t=1 (argument)
- Event B stored at timestamp t=2 (departure)
- THEN computed as:  $t_B - t_A = +1$ 
- Retrieval from time-indexed storage

**Part C (PEF-Based System):**
- Within PEF boundary (continuous present):
  - Reconstruct Event A from echo-traces → [argument + WE role + past
marker]
  - Reconstruct Event B from echo-traces → [departure + WE role + past
marker]
  - Apply THEN as relational operator: evaluate B as downstream from A
  - WE composite role maintained across both reconstructions
- No timestamps shown
- Annotation: "Both events reconstructed in present; relation computed
without timeline"

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Figure 5 - Memory Reconstruction Process Flow

Figure 5 shows a detailed process flow for memory reconstruction:

- Input: Query about past event
- ↓
- Remains in PEF (present state)
- ↓
- Access Echo-Trace Memory System
- ↓
- Retrieve relevant traces (patterns, not records)
- ↓
- Reconstruction Module assembles representation
- ↓
- Apply current context constraints
- ↓
- Mark as "reconstructed past content"
- ↓
- Evaluate within present frame
- ↓
- Output: Past event representation (constructed, not retrieved)
- All steps shown within PEF boundary
- Annotation: "System never leaves present state"

END OF SPECIFICATION